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Anomaly Detection For Vietnamese Financial Market

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***Abstract*—In the ever-evolving financial market landscape, the detection and forecasting of anomalies play a pivotal role in investment decision-making and risk management. This study focuses on identifying anomalies within the VN30-Index of the Vietnamese stock market. The continuous fluctuations in stock data pose significant challenges for investors seeking to make accurate decisions. We address this issue by leveraging foundation models (FMs) in time series analysis, particularly utilizing the MOIRAI model based on the Transformer architecture. These models, pre-trained on large datasets, exhibit the capability to handle diverse and complex data efficiently. Our proposed methodology for detecting anomalies is based on probabilistic forecasting and comprises two main stages: fine-tuning the foundation model on VN30 data and calculating the deviations to identify anomalies. The results indicate that the fine-tuned model significantly improves forecast accuracy and successfully identifies critical anomalies in the VN30-Index. These anomalies highlight substantial trend shifts and can inform strategic investment decisions. Compared to traditional strategies like Bollinger Bands and Moving Average Crossover (MAC), our anomaly-based investment strategy demonstrates promising outcomes. This research underscores the potential of foundation models and probabilistic forecasting methods in supporting investment decisions, marking a novel approach for optimizing investment returns and risk management in stock markets.**

***Index Terms*—Anomaly Detection, Time Series Forecasting, Financial Time Series, Transformer, Foundation Models, Deep Learning, Investment Strategy.**

I. INTRODUCTION

Detecting and forecasting anomalies in financial markets is crucial for informed investment decision-making and risk management. Anomaly detection identifies points where time series statistical properties shift, signaling market condition changes that may require strategic responses. Traditional methods like CUSUM and Bayesian approaches, while effective to some extent, often struggle with financial time series non-linearities and complexities. Given the high volatility and unpredictability of financial data, there is a growing need for advanced approaches to accurately model and detect these complex patterns.

Recent advances in deep learning, particularly the advent of Transformer [1] architectures, have opened new avenues for time series analysis. Transformers, initially designed for natural language processing tasks, have demonstrated exceptional performance in handling sequential data due to their ability to model long-range dependencies. This capability has spurred interest in applying Transformers for financial time series forecasting and anomaly detection.

Foundation models, which are pre-trained on large and diverse datasets, embody generalized knowledge that can be fine-tuned for specific tasks. The MOIRAI [2] model, based on the Transformer architecture, exemplifies such advancements by leveraging its pre-trained capabilities to handle complex and heterogeneous time series data efficiently. In our study, we propose leveraging the MOIRAI model to detect anomalies within the VN30-Index, a major index of the Vietnamese stock market.

Our methodology involves two primary stages:

- Fine-tuning the foundation model for time series data in forecasting
- Identifying anomalies through probabilistic forecasting

By addressing the limitations of traditional anomaly detection methods and harnessing the power of foundation models, our approach aims to provide a robust framework for anomaly detection in time series data. This research not only underscores the potential of deep learning models in time series analysis but also offers practical insights for enhancing investment strategies and risk management in the Vietnamese financial market.

II. RELATED WORK

The evolution of time series modeling has seen significant advancements, moving from traditional approaches to

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more sophisticated, data-driven methods. Early models such as Recurrent Neural Networks (RNN) [3] and Long Short-Term Memory (LSTM) [4] networks laid the groundwork for sequence modeling but faced limitations in capturing long-term dependencies due to issues like vanishing or exploding gradients. The introduction of Transformer architectures marked a pivotal shift in the field. Their self-attention mechanism enabled parallel processing of sequence data and effective management of long-range dependencies, expanding applications beyond natural language processing to time series forecasting. This innovation addressed key limitations of RNN and LSTM, particularly in handling varying sequence lengths and long-term patterns.

Building upon the success of Transformers, the advancement of foundational models, such as TimesFM [5], Lag-Llama [6], Chronos [7], MOMENT [8], MOIRAI and others, has significantly bolstered time series analysis. These models offer the advantage of encapsulating generalized knowledge that can be fine-tuned for specific tasks, providing robust solutions for complex and heterogeneous time series data. Among recent innovations, the MOIRAI model stands out for its integration of probabilistic forecasting within a Transformer-based framework. Unlike deterministic models, MOIRAI generates predictions as probability distributions, enabling the estimation of confidence intervals. This approach enhances uncertainty quantification, crucial for informed decision-making in various domains of time series forecasting.

The progression from traditional models to Transformer-based foundation models with probabilistic capabilities, as exemplified by MOIRAI, represents a significant leap in time series analysis. This evolution offers improved performance and efficiency, particularly in capturing complex patterns and uncertainties inherent in temporal data.

III. Methodology

Our proposed approach for detecting anomalies in time series data is based on probabilistic forecasting. The method leverages the power of foundation models that have been pre-trained on large datasets and are capable of zero-shot performance on unseen data. Our process consists of two main phases: fine-tuning a probabilistic forecasting model and calculating distances to identify anomalies.

A. Phase 1: Fine-tuning the forecasting foundation model

In this study, we employ the MOIRAI model, a state-of-the-art transformer-based time series forecasting model, but have Encoder only, as our foundational model for anomaly detection. The MOIRAI model was chosen due to its advanced capabilities in handling multivariate time series data with dynamic patch size projections and robust attention mechanisms. Notably, MOIRAI supports processing multivariate time series through its multi-patch size projection, which allows it to handle varying frequencies by splitting the input time series into non-overlapping patches of different sizes. Additionally, the Any-Variate Attention mechanism enhances the model's ability to simultaneously handle multiple variables by using Rotary Positional Embeddings (RoPE) [9] and *Binary Attention Bias* [10]. This mechanism ensures that the model captures the positional information while maintaining permutation equivariance and invariance, crucial for learning dependencies in time series data. The strength of MOIRAI lies in its probabilistic nature, predicting distribution parameters rather than single-point estimates. This approach enables uncertainty assessment, with broader confidence intervals reflecting higher uncertainty. As a foundational model, MOIRAI forecasts across diverse data domains, learning parameters of a mixture distribution to handle various data types effectively. The component distributions of the mixture include the Student's t-distribution, Log-normal distribution, Negative binomial distribution [11], and Low variance normal distribution. The mixture distribution with c component of MOIRAI has a probability density function defined as follows (1):

$$p(\mathbf{Y}_{t:t+h}|\hat{\phi}) = \sum_{i=1}^{c} w_i p_i(\mathbf{Y}_{t:t+h}|\hat{\phi}_i) \tag{1}$$

where $\mathbf{Y}_{t:t+h} = (\boldsymbol{y}_t, \ldots, \boldsymbol{y}_{t+h-1})$ represents the forecasted time horizon with a prediction length of h starting at time t, $\hat{\phi} = \{w_1, \hat{\phi}_i, ..., w_c, \hat{\phi}_c\}$ and p_i is the i^{th} component distribution of the probability density function.

To fine-tune the MOIRAI model for our specific dataset, we freeze the majority of the model layers and only update the weights of the final three components located in the last three layers of the model, including one fully connected layer and two normalization layers. This approach minimizes the risk of overfitting, given the vast number of parameters in the MOIRAI model while allowing the model to adapt to the specific characteristics of our financial time series data.

B. Phase 2: Distance Calculation and Anomaly Detection

At each time step t, we employ MOIRAI to sample a set of n samples $\hat{y}_i(t)$, which represent n potential outcomes at time step t. We then calculate the deviation of these n predicted values from the actual value y_t at time step t using the Mean Absolute Error (MAE), defined as follows (2):

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^{n} |y_t - \hat{y}_i(t)| \tag{2}$$

After the computational process, we obtain a set D of deviation values. We proceed to identify the top K anomalous points as follows: Instead of assigning a fixed value to K, we determine K by selecting the top $P\%$ of the largest distances in the set D. The parameter P can be flexibly determined based on user experimentation, a lower P value may result in identifying points with a higher likelihood of being anomalous. In summary, the value of K is determined as follows (3). The anomaly detection process can be summarized as Algorithm 1.

$$K = \left\lceil \frac{P}{100} \times |D| \right\rceil \tag{3}$$

Algorithm 1 Anomaly Detection Algorithm

Require: Time series data X, fine-tuned MOIRAI model M, threshold P

$W \leftarrow$ Create rolling prediction window from X
$D \leftarrow$ Init distance list
for each window w in W **do**
    $\hat{y} \leftarrow M(w)$ ▷ Make Prediction
    $d \leftarrow \text{MAE}(w, \hat{y})$ ▷ Distance Calculate
    Append d into D
end for
Determine K using the threshold P
$T \leftarrow$ top K largest distance in D
return T as anomalous list

IV. Experiments

In this section, we detail the experimental setup and results of applying our proposed anomaly detection methodology to the Vietnamese stock market, particularly focusing on the VN30-Index. This involves the collection and preprocessing of data, the fine-tuning of the MOIRAI model, the calculation of deviations, and the subsequent identification of anomalies. We further compare the performance of our method against traditional strategies like Bollinger Bands and Moving Average Crossover.

A. Dataset Information

The dataset comprises daily closing prices of 30 VN30 basket stocks and the VN30-Index from January 1, 2020, to January 1, 2024, totaling 991 trading days. Each column represents a stock's closing price, with the VN30-Index in the final column. Non-trading days were excluded to ensure continuity. Z-score normalization was applied to the data prior to model input, enhancing performance and accelerating training convergence. This preprocessing step standardizes the data, facilitating more effective learning across different stock scales.

B. Fine-tuning the MOIRAI model

In this stage, we use the MOIRAI model, a foundation model based on the Transformer architecture, pre-trained on a diverse range of time series data. We select the MOIRAI-Base variant, which consists of 91 million parameters across 12 layers, with $d_{\text{model}} = 768$, $d_{\text{ff}} = 3072$, and 12 attention heads.

1) Training and Validation Data: The dataset, encompassing 991 trading days, was partitioned into training and validation subsets. The initial 900 days were designated for model training, with the subsequent 91 days allocated for validation purposes. To assess the model's predictive capabilities, an independent test dataset was acquired, comprising 91 trading days commencing from January 2, 2024. This temporal segregation of data ensures a robust evaluation of the model's generalization performance on unseen market conditions.

2) Model Freezing Strategy: Given the relatively small size of our dataset compared to the model's 91 million parameters, we took measures to prevent overfitting, as detailed in Section III-A. Specifically, we froze most of the model's layers, reducing the number of trainable parameters to 5.2 million for the training process.

3) Fine-tuning Procedure: The fine-tuning process was conducted to adapt the model to the specific characteristics of the VN30 stock data. In this configuration, the 30 constituent stocks of the VN30 were utilized as covariates, while the VN30-Index served as the target variable. The optimization was performed using the AdamW [12] algorithm with the following hyperparameters, $\text{lr} = 0.005, \text{weight\_decay} = 0.5, \beta_1 = 0.9, \beta_2 = 0.98$. An early stopping mechanism was implemented to mitigate overfitting, terminating the training process if the validation loss failed to improve for 5 consecutive epochs. The model exhibiting the lowest validation loss was subsequently selected for further analysis.

C. Deviation Calculation and Anomaly Detection

Following fine-tuning, the model was employed to generate probabilistic forecasts. For each time step, we performed 100 samples from the model's predictive mixture distribution and calculated the distances as outlined in Section III-B.

For the test dataset comprising 91 trading days, we used the fine-tuned model to generate forecasts starting from February 13, 2024. The anomalies were identified by selecting the top K largest deviations in MAE between the forecasted and actual values. In our experiments, we investigated anomalies by setting $K = 5$ and $K = 10$, corresponding to $P = 10\%$ and 20%, respectively.

D. Experimental Results

1) Model Performance Before and After Fine-tuning: The impact of fine-tuning on the model's performance was evaluated using several metrics, including Mean Squared Error (MSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and Dynamic Time Warping (DTW). Table I summarizes the improvements across these metrics. Since MOIRAI is a probabilistic forecasting model, the evaluation results were averaged over five runs on the test set. To further ensure the objectivity of the evaluation process, we also compared the performance of the model with the AutoArima [13]. We use the AutoArima package to automate the process of fitting ARIMA models by selecting the optimal model parameters based on statistical criteria.

Results in Table I demonstrate significant improvement in all performance metrics after model fine-tuning. MSE, MAE, and MAPE show substantial reductions, indicating decreased prediction errors. The lowered DTW metric suggests enhanced capture of temporal dynamics. These findings indicate that fine-tuning has improved the model's overall performance, resulting in more accurate and reliable predictions.

2) Visualization of Forecast Performance: Fig. 1 illustrates the predictive capability of MOIRAI-Base prior to fine-tuning

TABLE I
COMPARISON OF MOIRAI-BASE MODEL PERFORMANCE BEFORE AND AFTER FINE-TUNING AGAINST BASELINE AUTOARIMA

Metric	MOIRAI-Base		Baseline
	Before	*After*	*AutoArima*
MSE	0.0843	**0.0296**	0.075
MAE	0.2668	**0.1451**	0.2167
MAPE	0.5626	**0.3247**	0.6134
DTW	11.4202	**5.8591**	8.1015

on the VN30 dataset. These values exhibit substantial deviations from actual values, and the wide confidence intervals indicate a high degree of uncertainty in the model's predictions. Conversely, Fig. 2 demonstrates reduced discrepancies between predicted and actual values, coupled with narrower confidence intervals, indicating the enhanced predictive performance of the model post-fine-tuning. This comparative analysis underscores the significant improvement in predictive accuracy and reliability achieved through the fine-tuning process, highlighting the model's adaptability to the specific characteristics of the VN30 index.

3) Anomaly Identification Results: Fig. 3 and Fig. 4 illustrate the identification of the top 5 and top 10 anomalies in the VN30-Index during the test period. Our method can readily detect points that deviate significantly from the average data shape (areas circled in red), while also identifying instances where the VN30-Index begins to exhibit upward or downward trends. These findings present a potentially valuable insight that may serve as a reference for investors seeking a basis to support decision-making in securities trading within the Vietnamese market, particularly concerning the VN30-Index trajectory. The results suggest a promising approach that could be utilized to inform investment strategies and market analysis in the context of Vietnam's stock market dynamics.

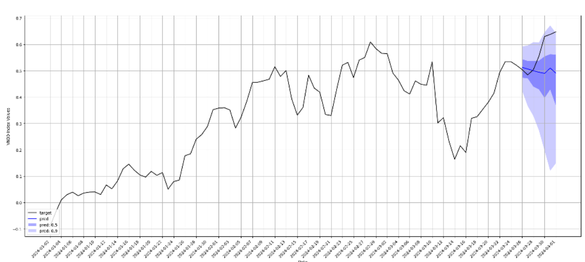

Fig. 1. Forecast performance of MOIRAI-Base before fine-tuning

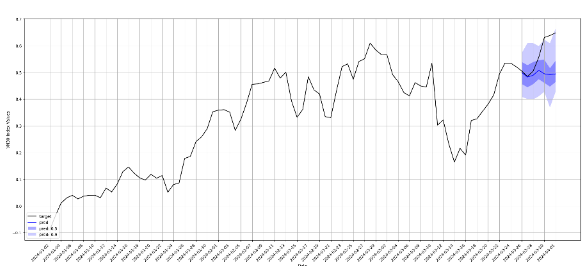

Fig. 2. Forecast performance of MOIRAI-Base after fine-tuning

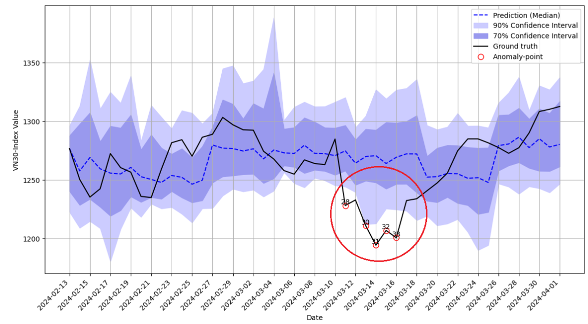

Fig. 3. Top-5 anomalies identified on the test dataset

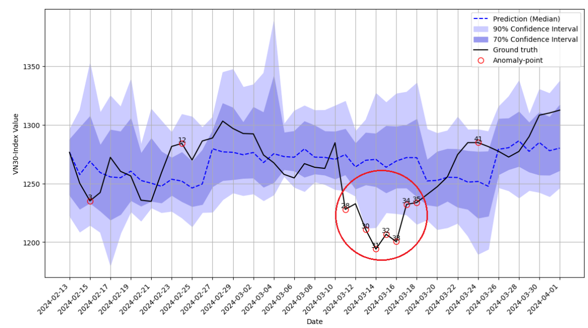

Fig. 4. Top-10 anomalies identified on the test dataset

E. Investment Strategy Based on Anomalies

To evaluate the practical utility of our anomaly detection method, we devised two investment strategies: One combining Moving Average Crossover with anomaly detection, and another based on buying at identified anomalies outside the 90% confidence interval and selling the following day.

1) Strategy 1: Combining Moving Average Crossover (MAC) Strategy and Anomaly Detection: Investors execute buy and sell actions according to the MAC signals and maintain their positions until encountering an anomaly point, at which point they will immediately adjust their position and exit the market. We use the dataset of 250 trading days in 2019 to generate experience of this strategy. The dataset was split into five 50-day subsets, and the investment strategy was applied over 30 days in each subset. Table II presents the results, showing a Sharpe Ratio and cumulative returns for each subset, both with and without anomaly detection.

The Sharpe Ratio is a metric used to assess the performance of an investment fund or a stock. A higher Sharpe Ratio signifies that the investment provides a higher return relative to the risk it incurs. Conversely, a lower Sharpe Ratio indicates an inadequate benefit-to-risk ratio. This metric is defined by the following (4):

TABLE II
INVESTMENT STRATEGY COMPARISON: MAC VS. MAC WITH ANOMALY DETECTION

Data subset	Dataset 1	Dataset 2	Dataset 3	Dataset 4	Dataset 5
Moving Average Crossover					
Sharpe Ratio	2.8237	-4.3702	-2.7379	-4.4191	-3.1207
Return (%)	0.02734	-0.0387	-0.0134	-0.0023	-0.01962
Number of Trades	3	4	4	3	3
Moving Average Crossover with Anomaly Detection					
Sharpe Ratio	**2.8748**	**-3.1490**	**3.1425**	**1.8918**	-3.1207
Return (%)	**0.0464**	**-0.0168**	**0.0142**	**0.0032**	-0.01962
Number of Trades	**2**	**2**	4	**2**	**2**

TABLE III
INVESTMENT STRATEGY COMPARISON: BOLLINGER BANDS VS. ANOMALY-BASED

Strategy	Sharpe Ratio	Cumulative Return (%)
BB (Window 20)	-1.3008	-0.0225
BB (Window 40)	0.2524	0.01767
BB (Window 60)	0.2752	0.01768
BB (Window 80)	0.3074	0.0178
BB (Window 100)	-0.9358	-0.0082
Anomaly-based	**0.6065**	**0.0311**

$$\text{Sharpe Ratio} = \frac{R_p - R_f}{\sigma_p} \quad (4)$$

where R_p represents the expected return of the investment (mean), R_f is the risk-free rate (typically the rate on government bonds or short-term deposits), and σ_p is the standard deviation of the investment, reflecting its level of volatility.

2) Strategy 2: Anomaly-based Buy and Sell: In this strategy, an investor buys stocks at identified anomalies that lie outside the 90% confidence interval and sells them the next trading day. The strategy's performance was compared against the traditional Bollinger Bands (BB) method over 250 trading days in 2019. The results, depicted in Table III, show the Sharpe Ratio and cumulative return percentages.

Our findings indicate that trading strategies incorporating anomaly detection outperform conventional approaches such as Bollinger Bands or Moving Average Crossover. The results suggest that the identification and utilization of anomaly points can significantly enhance the efficacy of investment strategies in comparison to traditional technical analysis methods.

V. CONCLUSION

Our study demonstrates the effectiveness of foundation models for time series data in financial analysis, particularly in forecasting and anomaly detection. The results provide valuable insights into market dynamics, potentially enhancing investment decision-making processes. This highlights the success of our approach and the importance of anomalies in stock market investments.

Time series-specific foundation models have only recently seen significant development, resulting in limited reference materials and challenges in effective fine-tuning. Future work will focus on enhancing the forecasting phase of this method, exploring more sophisticated distance metrics, and implementing advanced model fine-tuning techniques. Specifically, we plan to investigate the application of LoRA [14] for more efficient fine-tuning, as an alternative to the current model freezing approach. This research contributes to the growing body of knowledge in financial time series analysis and provides a new approach to anomaly detection in time series data for future studies.

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